Tutorial 4: K-means and Gaussian Mixtures

From hard assignments to probabilistic clustering

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How to use this tutorial

- Attempt each problem before reading its solution.
- Write the main calculation or argument clearly.
- Explain what the result means, not only the final number or formula.
- Complete Tasks 1–3 first. The final task is an optional challenge.

Task 1: K-means initialization and k-means++

Consider one-dimensional data:

$$x = \{1, 2, 5, 8, 9\}.$$

We want $K = 2$ clusters.
First use initialization A:

$$\mu_1 = 1, \qquad \mu_2 = 5.$$

1. Perform one assignment step.
2. Compute the initial objective.
3. Update the centroids.

Then consider initialization B:

$$\mu_1 = 2, \qquad \mu_2 = 8.$$

Repeat the same calculation.
Compare the two initializations.

Task 2: GMM responsibility calculation

A point x may be generated by three Gaussian components.
The component information is:

$$\pi_1 = 0.2, \qquad p(x|z=1) = 0.9,$$

$$\pi_2 = 0.5, \qquad p(x|z=2) = 0.3,$$

$$\pi_3 = 0.3, \qquad p(x|z=3) = 0.4.$$

1. Compute the responsibility of each component.
2. Which component has the largest posterior probability?
3. Explain why the largest π_k does not always produce the largest responsibility.

Task 3: EM weighted parameter update

After the E-step, one component receives the following responsibilities:

$$\gamma = (0.1, 0.7, 0.8, 0.4)$$

for observations:

$$x = (2, 4, 7, 9).$$

1. Compute the effective membership:

$$N_k = \sum_i \gamma_i.$$

2. Compute the updated mean.
3. Compute the updated mixing weight assuming $n = 4$.
4. Explain why EM uses weighted statistics.

Optional challenge: implement clustering with GMM

Use a synthetic dataset and compare K-means with Gaussian mixture models.

1. Generate a two-dimensional dataset containing several groups.
2. Train a K-means model and a GMM model.
3. Visualize the clustering results.
4. Compare the hard assignments from K-means with the soft responsibilities from GMM.

Tutorial 4 complete

Questions and discussion